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Research Paper

Relationship between peer review quality and scientific impact: Insights from LLMs-assessed reviews

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ABSTRACT

The significant rise in scientific publications has made pre-publication peer review crucial for safeguarding research quality and enhancing scientific impact. Yet, correlation between peer review quality, indicative of a rigorous peer review procedure, and the subsequent scientific impact is rarely investigated. Here, this study leverages Google's large language model (LLM) named Gemini 2.0 Flash along with the few-shot prompt to automatically evaluate peer review quality, and investigate the correlation between the automatically assessed peer review quality (indicated by review scores) and scientific impact (measured by post-publication citations). First, the subsample reliability analysis reveals excellent agreement between review scores evaluated by humans and those generated by Gemini 2.0 Flash, underscoring the potential of LLMs in automatically assessing peer review quality. Then, empirical analysis indicates a positive association between automatically assessed review scores and citations. Specifically, a one-point increase in the review score is associated with an average increase of approximately five citations (coefficient = 5.440, $p < 0.01$). This positive effect remains robust in several robustness analyses, including transforming citation counts into logarithmic form, excluding papers within the top 1% of citations, and applying a negative binomial regression model. In addition, all analyses remain robust when using the average and "year-normalised" citations as dependent variables, when using first-author information to proxy control variables, and when employing a zero-shot prompt for Gemini 2.0 Flash responses, despite the relatively lower estimated effect. These findings highlight the importance of peer review quality in enhancing the scientific impact and the potential of using LLMs to automatically assess the review quality. Despite the promise, the widespread adoption of LLMs in automatically assessing peer review relevant tasks warrants further investigation.

1. Introduction

Peer review has long been regarded as a rigorous scientific stage for research evaluation (Bornmann, 2011). By offering detailed and critical feedbacks from experienced researchers in addressing several aspects of a manuscript (such as the methodological rigor, result interpretations, and presentation), quality of the manuscript can be significantly enhanced (Mulligan et al., 2013). This, in turn, may influences the post-publication scientific impact (Abramo et al., 2024; Kang et al., 2025; Sun et al., 2024; Tennant & Ross-Hellauer, 2020). However, the exponential growth of publications has also led to a proliferation of low quality studies, often attributed to the failure of peer review (Bornmann & Mutz, 2015; Noorden, 2025; Van Noorden, 2023). In this context, assessing the

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quality of peer reviews becomes imperative. As a higher review quality score is typically interpreted as evidence that a manuscript has undergone rigorous evaluation, thereby improving the overall quality and contributing to a higher scientific impact. Despite this, existing studies measure review quality often rely on self-assigned scores from reviewers, which is subject to individual bias and may lead to biased estimations (Abramo et al., 2024; Patterson & Harris, 2009). As a result, examining the relationship between peer review quality and scientific impact still warrants in-depth exploration.

Distinguishing from the self-evaluated process of peer review quality scores (Wu et al., 2024), the review quality instrument (RQI) has been well-developed to provide a more objective, accountable, and transparent assessment, particularly for peer review in life sciences and biomedical domain. Those instruments are used to evaluate whether reviewers effectively fulfill their responsibilities, including critically evaluating submissions and providing constructive feedback on several items to enhance research quality. Previous studies have employed RQIs to score the review quality (Van Rooyen et al., 1999); however, these efforts are often limited by small sample sizes due to the time-intensive nature of manual evaluation. Additionally, human assessments are susceptible to fatigue, subjectivity, and bias, which can compromise the reliability of the results. The recent emergence of generative AI, particularly the large language models (LLMs), which demonstrate proficiency across a wide range of tasks and offer a promising solution to these challenges (Sun, 2025).

LLMs have demonstrated significant capabilities in performing various research tasks, offering transformative potential for the traditional scientific research paradigm (Grossmann et al., 2023; Wei et al., 2022). Recent studies have shown that LLMs, such as OpenAI's ChatGPT-4, can assist in pre-peer review processes by functioning as review checklist assistants (Goldberg et al., 2024). They have also shown promise in generating peer review comments that address key aspects of peer review process, including recognizing the strengths (Drori & Te'eni, 2024), weaknesses (Widhawati et al., 2024), clarity (Liang et al., 2024), novelty (Kadi & Aslaner, 2024; Tyser et al., 2024), and reproducibility (Srivastava, 2023) of a submitted paper. These capabilities have been demonstrated across multiple LLM models, including LLaMA2, PaLM2, among others. Moreover, LLMs have been shown to provide helpfulness ratings, generate recommendation scores, and even produce final decisions for submitted papers (Robertson, 2023; Saad et al., 2024; Mike Thelwall & Yaghi, 2024). Continuous advancements have improved the ability of LLMs to generate feedback that closely aligns with human reviews, offering more specific and constructive feedback on scientific papers (D'Arcy et al., 2024). These improvements highlight the potential of LLMs to streamline and enhance peer review tasks, ultimately contributing to the automatic peer review process (Thelwall, 2024b).

Further evidence suggests that LLMs can provide feedback on peer reviews that is comparable to human evaluators (Liang et al., 2024). However, the research utilizing ChatGPT-4 have not fully integrated standardized RQIs, which are essential for evaluating key aspects of peer review quality, even though some aspects have been considered. Moreover, several leading LLMs, including ChatGPT-4, Claude (version 3.5 Sonnet), and Gemini (versions 1 and 1.5), have exhibited signs of declining cognitive performance, as their earlier versions outperformed more recent counterparts (Dayan et al., 2024). These concerns, coupled with the rapid pace of LLM version development, underscore the need for newer models with enhanced contextual understanding and processing capabilities. Google's latest LLM versions have demonstrated superior performance (Thelwall, 2024a), consistently ranking first on the Hugging Face Chatbot Arena Leaderboard[1]. This positions them as strong candidates for the automated evaluation of peer review quality.

In this study, I employ Google's most advanced LLM, Gemini 2.0 Flash, to automate the evaluation of peer review quality using the most widely recognized RQI tool. By incorporating those tools, this study aims to enhance the rigor of the evaluation process. Given that RQIs are predominantly applied in the biomedical research domain, my analysis focuses on this area, utilizing open peer review comments retrieved from *eLife*, a leading journal in the biomedical field. Specifically, the relationship between peer review quality, as automatically assessed by Gemini 2.0 Flash, and scientific impact, measured through post-publication citations is investigated. The objectives of this study are to explore:

1). The reliability of LLM as a tool for assessing peer review quality;
2). The correlation between LLM-measured peer review quality and the impact of publications.

This study makes contribution in advancing the understanding of how the peer review quality influences scientific impact of academic research and demonstrates the potential application of LLMs in enhancing the assessment of review quality.

2. Related work

2.1. Reviewers' role

Serving as gatekeepers, peer reviewers hold considerable influence over the peer review process (Kang et al., 2025). The extent to which they maintain objectivity to ensure an unbiased peer review is still the focus that attract wide scientific attention (Lee et al., 2013). One of such research domains utilized rich dataset attributable to the open peer review of investigating the reviewers' role (Sun et al., 2022). A key focus of the extant research is to assess reviewer performance through analysis of the review scores they assigned, treating these ratings as indicators of their perceived research "quality" and explore the relationship between those scores and the subsequent scientific impact (Bornmann, 2017; Kousha & Thelwall, 2024).

[1] The website is accessed via: https://huggingface.co/spaces/lmsys/chatbot-arena-leaderboard.

For example, by using the data from Italian research assessment exercise (RAE), Abramo et al. (2024) found that papers with the highest evaluation scores demonstrated significantly higher citations, whereas lower-scored papers showed no such correlation with citation impact. This indicates that reviewers possess greater accuracy in identifying excellent papers. Similarly, Patterson & Harris (2009) conducted an investigation using reviewer scores from manuscripts submitted to Physics in Medicine and Biology, discovering that publications classified in the highest quality category received approximately double the citations. These findings demonstrate that reviewer scores correlate with citation impact primarily for high-quality papers, while other studies present evidence showing that manuscripts receiving lower peer reviewers consistently attract relatively fewer citations when published. For instance, study by Siler et al (2015) demonstrated this relationship using comprehensive data from both published and rejected manuscripts across three elite medical journals. In the same vein, numerous studies across diverse research domains have employed publicly available datasets to demonstrate consistent positive correlations between higher reviewer scores and increased citations (Bornmann, Marx, Schier, Thor, & Daniel, 2010).

Despite those positive findings, contrasting evidence has emerged from other investigations, which report that manuscripts with greater inconsistency in review scores subsequently garnered more citations (Xie et al., 2024), or higher citations or no significant association between review scores and eventual citation performance (Bartneck, 2017). A potential reason for the inconsistent findings may lie in the fact that the existing review scores are self-assessed by the reviewers themselves. These scores represent a perceived measure of research quality and only reflects the quality of the review itself. Moreover, such measurement is subject to inherent limitations, including subjective issues, inconsistent among reviewers (Bharti et al., 2022), and a tendency for junior reviewers to assign lower scores (Wu et al., 2024). In this regard, a more precise and objective approach to evaluating reviewer performance, for example, by employing RQI in a more objective and accountable manner is needed. However, the traditional RQI evaluation process is dependent on humans, suffers from human errors, and is time-consuming.

2.2. LLMs for peer review

To address the aforementioned challenges in evaluating peer review quality more effectively and objectively, LLMs demonstrate considerable promise. LLMs are the emerging artificial intelligence technologies trained on vast amounts of data from multiple sources to generate human-like responses. Since the first widely released LLM (ChatGPT) by OpenAI in 2022, numerous LLMs have been developed by various companies and institutions to accomplish multiple tasks in both industry and academia (Clusmann et al., 2023; Luo et al., 2025). They have showed promised in peer review relevant tasks, such as being used as checklist assistants during the pre-peer review stage to aid editors-in-chief in checking whether submitted manuscripts align well with the submission guidelines and academic standards of journals. In this type of task, existing literature demonstrated that LLMs exhibited potential in identifying errors in submissions and achieving higher accuracy in confirming submission checklists (Goldberg et al., 2024; Liu & Shah, 2023).

Notably, considerable research attention has centered on the capacity of LLMs aiding in reshaping the peer review process. Multiple investigations have examined potential LLM applications in simulating the dynamic and iterative nature of peer review and in automated review generation and assign review scores (Liang et al., 2024; Tan et al., 2024). LLMs such as GPT-3.5, PaLM2, and LLaMA2 have been demonstrated to outperform humans in generating reviews (Santu et al., 2024), succeed in offering critical comments (Tunga & Çelik, 2024), and provide sentiment analysis to predict paper decisions (Srivastava, 2023). Moreover, LLMs have shown the ability to produce weak to moderately strong predictions of peer review scores (Thelwall & Yaghi, 2024), and have been observed to be efficient in detecting gender and geographic bias in peer review (Sebo, 2024). Despite the significant potential of LLMs in aiding peer review, concerns regarding inherent biases and ethical risks emphasize that LLMs should serve as supportive tools rather than substitutes for human evaluators. The validation of LLM-generated results by humans and well-designed prompts are essential to ensure the quality and accuracy of LLM outputs (Huang et al., 2025).

In summary, given the inherent flaws of employing self-assessed review scores as proxies for peer review quality, as well as the potential of LLMs when compared with human evaluations to promote more effective peer review quality assessment through well-developed RQI, this study aims to extend this research direction by exploring the potential of LLMs in aiding peer review quality assessment tasks, and ultimately presenting a more precise exploration of the relationship between peer review quality and scientific impact.

3. Data and methods

3.1. Data

Peer review reports: Peer review reports (PRRs) for papers were downloaded from the website of the *eLife* journal, covering the time period from 2012 that the journal's initiation of publicizing PRRs up to 2021. I choose *eLife* as the target journal for the reasons that: 1) it advocates single-blind peer review policy by provides unprecedented access to complete PRRs; and 2) it is a well-known journal supported by prestigious organizations like the Howard Hughes Medical Institute (HHMI). PRRs published 2022 and after were excluded due to the announcement of a significant transformation in *eLife*'s peer review system, wherein the journal ceased making accept-or-reject decisions by traditional peer review, instead designating submissions as reviewed preprints (Else, 2022). As this change impacts the peer review process, PRRs from this period were not considered in the analysis. Additionally, only PRRs that included reviewers' information were selected, as reviewer identities may influence peer review quality and thus need to be accounted for (McNutt et al., 1990). After excluding PRRs without reviewer information, a total of 2,580 papers, along with their corresponding PRRs (one entire review per paper), were collected for the study.

Review quality instrument: In biomedical research domain, validated RQIs have been developed and widely utilized to enhance the peer review process in well-conducted trials by assessing the quality of PPR. According to the latest review by Superchi et al. (2019), a total of 24 RQIs were identified. Following an in-depth screening of these instruments, those designed to evaluate quality domains using Likert scales, with multiple items scored from 1 (indicating poor performance) to 4 or 5 (indicating excellent quality), were considered suitable for quantitatively assessing review quality. Based on these selection criteria, the most prominent RQI, which incorporates seven items for assessing reviews, was used in this study (Van Rooyen et al., 1999).

As presented in Table 1, seven items are designed to evaluate various aspects of the reviewer's performance, which better align with the reviewer instructions *eLife* provide for reviewer to write comments[2]. Specifically, the RQI assesses whether the reviewer has: (1) addressed the importance of the reviewed manuscript; (2) highlighted its originality; (3) identified the strengths and weaknesses of the methodology in the reviewed manuscript; (4) evaluated the presentation of the paper, including its writing and organization; (5) provided comments relevant to the result interpretation; (6) provided constructive comments; and (7) provided substantiated comments. Each item is scored based on a 5-point Likert scale, and the total score is calculated as the mean of the scores across the seven items.

Scientific impact: The number of citations a paper receives is widely used to proxy its scientific impact (Wolfram et al., 2021). Generally, the more a paper is cited, the greater its presumed scientific influence. For the 2,580 collected papers, citation counts were retrieved using the Web of Science (WoS) database. By utilizing the digital object identifier (DOI) of each paper, three metrics reflecting the post-publication scientific impact were obtained and calculated: the number of citations within the first three years after publication, the average citations within this period, and the "year-normalised" citations. The normalisation was performed using the average citations received by all "biology" publications within the three years following their publication.

3.2. LLMs and prompt development

LLMs: The LLMs utilized was Gemini 2.0 Flash (Google, 2024), the latest version developed by Google. Gemini 2.0 Flash has demonstrated exceptional performance in answering complex questions and providing detailed explanations of its "thinking" process. This capability to articulate its reasoning makes it particularly well-suited for tasks involving peer review quality assessment. In this study, the systematic evaluation of each item in the RQI requires not only accurate judgments but also a clear rationale for those judgments. The model's ability to simulate a structured decision-making process aligns well with the critical thinking required to assess the review quality effectively.

Few-Shot Prompt: Output performance of LLMs often depends on effective prompt engineering, to guide the model to mimic a natural conversational process. Specifically, I employed the widely used few-shot prompting technique to elicit Gemini 2.0 Flash's feedback for the review quality assessment task. Few-shot prompt provides contextual reasoning before assigning scores of each input PPR.

As the prompt template presented in Fig. 1, two important factors that align with the PPR context of papers in *eLife* were considered in constructing the few-shot prompt approach. First, the single-blind peer review practices employed by the *eLife* journal was simulated. Second, given the token limitations of Gemini 2.0 Flash as noted in existing work (Huang et al., 2025), I represented the original content of each paper using its abstract to capture the core manuscript content while having Gemini 2.0 Flash assume the role of a reviewer. Specifically, as shown in Fig. A1-A3, three examples representing high-, medium-, and low-quality review were used to guide Gemini 2.0 Flash in contextual reasoning.

Furthermore, three components that make up the benchmark prompt were included: 1) task description to clearly inform the LLM of Gemini 2.0 Flash specific task of review quality assessment (as presented in Fig. 1); 2) instructions of RQI items that providing Gemini 2.0 Flash a detailed explanation of the items in the RQI used to assess the quality of peer reviews (as presented in Fig. A4); and 3) query for task evaluation to specify the desired format of the output, including (a) scores for each RQI item as shown in Table 1, (b) a textual summary highlighting the strengths of the reviewed manuscript, (c) constructive comments for improvement, and (d) an explanation of the scores assigned by the model (as presented in Fig. A5).

The response output of the Gemini 2.0 Flash is structured in a well-organized JSON format, making it more suitable for further analysis. Considering that zero-shot prompt directly instructs Gemini 2.0 Flash to assess review quality without additional contextual information, the results were also included as a supplementary reference.

3.3. Reliability analysis and empirical strategy

LLMs results evaluation: To examine the validity of the results generated by Gemini 2.0 Flash, I conducted a reliability analysis involving two human evaluators who manually assessed the review quality of a subsample of 300 papers. Prior to the formal evaluation tasks, the evaluators underwent thorough training to ensure their consistency. To this end, I first make sure that they were familiar with the detailed explanations of each item for review quality assessments. Then a practice of evaluating the review quality by using 10 additional papers outside the sample was conducted and repeated to ensure they achieved excellent agreement before the formal evaluation.

Subsequently, they independently assessed the review quality of the subsample, and each item score they assessed were analyzed

[2] see: https://elife-rp.msubmit.net/cgi-bin/main.plex?form_type=display_rev_instructions#writing.

Table 1
RQI used for review quality assessment.

Items	Item explanation
1) Importance	The extent to which PPR highlight the importance of the reviewed manuscript to the research domain
2) Originality	The extent to which PPR emphasize the scientific originality and value of the reviewed manuscript
3) Methodology	The extent to which PPR underscore the strengths and weaknesses of the methodology of the reviewed manuscript
4) Presentation	The extent to which PPR stress the strengths and writing, organization and illustrations of the reviewed manuscript
5) Interpretation	The extent to which PPR provide comment regarding the result interpretation of the reviewed manuscript
6) Constructive comments	The extent to which PPR provide helpful and actionable comments and suggestions to improve the work
7) Substantiated comments	The extent to which PPR was with solid foundation that make the comment more credible and persuasive

Notes: The RQI form Van Rooyen et al. (1999) was used to assess the quality of PPR.

Prompt Used For Review Quality Assessment

Task description:
Please evaluate the following review using the 7-item Review Quality Instrument (RQI). Each item should be scored on a 5-point Likert scale (1=poor, 5=excellent).

Notification of single-blind peer review:
Note: This evaluation is conducted under single-blind review conditions, where the reviewer knew the author's identity but the author does not know the reviewer's identity.

Instruction of RQI items:
Evaluate these specific aspects:

Context:
Few-Shot Examples
Example 1: High-Quality Review
Example 2: Medium-Quality Review
Example 3: Low-Quality Review

Query for task evaluation:
Your Evaluation Task

Fig. 1. Structure of few-shot prompt used for review quality assessment.

using observed agreement, a common approach for assessing score consistency (Cohen, 1968). I employed observed agreement because of the narrow range of the Likert-scale scores (1 to 5), where a difference of 1 point may still represent substantial consistency. To consider this issue, I followed established literature recommendations (Cohen, 1968; Mehta et al., 2024), treating differences of ≤ 1 as essentially or nearly identical. Specifically, the observed agreement was calculated using the formula: observed agreement = (Number of agreements / Total observations), with the value between 0.40 and 0.75 indicates fair or good agreement, while a value above 0.75 implies excellent agreement (Fleiss et al., 2013).

Then, to evaluate whether the LLMs of Gemini 2.0 Flash exhibit performance comparable to that of human evaluators, as found previously (Liang et al., 2024), the consistency between human and LLM evaluations was also measured by using the observed agreement. Notably, as the primary interest of this study lies in the total score reflecting the overall performance of the LLMs, the observed agreement specifically for this indicator was calculated.

Relationship between review quality assessed by LLMs and scientific impact: A linear regression model was employed to control for a set of variables potentially related to scientific impact, as well as a year dummy variable to account for journal-specific factors that might influence academic impact of research. Specifically, the linear regression model is expressed as follows:

$$Y_{it} = \alpha + \beta Review_quality_{it} + \theta \boldsymbol{X}_i + \varphi_t + \varepsilon_{it}, \tag{1}$$

where Y_{it} represents the citation, reflecting the scientific impact of paper i published in year t. $Review_quality_{it}$ denotes the review scores (the mean of the scores across the seven items) automatically assessed by Gemini 2.0 Flash. Based on my previous research (Sun et al., 2023), $\boldsymbol{X}_i$ is a vector of control variables accounting for the variation of number of citations due to other factors, they are thus included for more precise estimates. These variables are: 1) the title length measure by the number of words in a paper's title (Deng, 2015); 2) the number of pages measured by calculating how many pages of a paper (Stremersch et al., 2007); 3) the length of abstract determined by the number of words for a paper's abstract (Rostami et al., 2014); 4) the number of references measured in terms of the references cited by a paper (Bornmann et al., 2012); 5) the number of authors determined by calculating how many authors for a paper (Bornmann et al., 2014); 6) the institution of last corresponding author measure by whether it is a top 100 university[3] (Kang et al., 2025; Sun et al., 2023); 7) the country of the last corresponding author quantified by whether it is the English speaking countries (Farshad et al., 2013); and 8) the academic status of reviewers determined by whether the h-index of reviewers is higher that of the last

[3] In accordance with the convention that the last corresponding author is typically the senior researcher primarily responsible for managing the progress of the work in the field of life and biomedical sciences, the information of the last corresponding author was thus used to construct the control variable.

corresponding author (Biscaro & Giupponi, 2014; Teplitskiy, et al., 2018). φ_t is the year fixed effect to control for and capture the time-specific factors that influence citation patterns in a given year, and ε_{it} is the error term. To ensure the robustness of the regression estimation, I also conducted several robustness checks, including transforming citation counts into logarithmic form, excluding papers within the top 1% of citations, and applying a negative binomial regression model.

4. Results

The results are presented in the following three steps. First, the reliability of Gemini 2.0 Flash in assessing review quality by comparing its performance with human evaluators through a subsample analysis is tested. Then, the statistical properties of the review quality scores generated by Gemini 2.0 in the dataset is presented. Lastly and importantly, the core analysis of exploring the relationship between review scores assessed by Gemini 2.0 and citations is conducted.

4.1. Reliability analysis of Gemini 2.0 Flash in evaluating review quality

The results of human evaluators and Gemini 2.0 Flash's output for the subsample are presented. Since each item in the RQI is scored on a scale of 1 to 5 by both human and Gemini 2.0 Flash evaluators, I calculate the average score for each item across the 300 subsample papers. As shown in Fig. 2, Gemini 2.0 Flash scored comparably to human evaluators on item 2 (importance of the study, as outlined in Table 1). This similarity is further reflected in the mean and standard deviation values of 3.51 (0.93) and 3.29 (1.14), respectively (Table 2), confirming that scores assigned by both evaluators were comparable. Scores for all remaining items demonstrated consistent patterns (Fig. 2 and Table 2).

To evaluate the reliability between the two evaluators, the agreement of the evaluated item scores is calculated. Results in Table 2 show that all item scores achieve observed agreement values are around 0.99, implying excellent agreement between the two evaluators.

To further validate the consistent performance of human evaluators and Gemini 2.0 Flash's output, the agreement results for the average overall score of the review are presented in Table 3[4]. The values are 0.86 and 0.79, suggesting that the performance of using Gemini 2.0 Flash for review quality assessment is comparable to the human evaluators. The excellent agreement is observed, further supporting the alignment between the human and automation evaluation of PPRs. For the Gemini 2.0 Flash constituency with two repetitions, an excellent agreement of 0.98 is also observed.

4.2. Statistics of review scores generated by Gemini 2.0 Flash

An analysis of the responses generated by Gemini 2.0 Flash is conducted, focusing on analyzing reviewers' roles in the peer review process to assess review quality (Gendler Szabó, 2001). As shown in Fig. 3, the score distribution generated by Gemini 2.0 Flash indicates that the scores for the seven items predominantly range between 3 and 5.

4.3. Relationship between review quality and citations

To examine the relationship between review scores assigned by Gemini 2.0 Flash and the scientific impact of a paper, measured by the number of citations within the first three years after publication, the results of the linear regression analysis are reported in Column 1 of Table 4. The estimated coefficient is 5.440 and is statistically significant at the 1% level, suggesting a positive association between a paper's review score and its subsequent scientific impact. Specifically, a one-point increase in the review score (reflecting higher-quality peer review) corresponds to an average increase of approximately five citations.

Several robustness checks are conducted to further validate the estimated effect. First, to address potential heteroscedasticity, the dependent variable (number of citations) is transformed into its logarithmic form using the formula: ln(citations). The results, present in Column 2 of Table 4, demonstrate consistency with the baseline estimation, as the coefficient remains positive and statistically significant. Then, to mitigate the influence of outliers, papers with an exceptionally high number of citations are excluded and re-estimate the benchmark model. Specifically, papers ranking in the top 1% by citations are removed, and the result are reported in Column 3 of Table 4. The coefficient for *review quality* remains positive, with the magnitude and significance consistent with those from the baseline model. Finally, as a further robustness check, I account for the count data nature of the number of citations by specifying a negative binomial regression model to address potential non-linearity in the data. The corresponding results are presented in Column 4 of Table 4. The signs, magnitudes, and statistical significance of the coefficients were similar to those in the baseline model, suggesting that the observed effect is robust across alternative model specifications.

In addition, several robustness checks were performed to confirm the robustness of the findings. First, when using both average citations and "year-normalised" citations as dependent variables, the findings shown in Table A2 were consistent with the main result displayed in Table 4. Second, when replicating the analytical process using a zero-shot prompt to automatically assess the review scores, the positive effect persisted (Figs. A6 and A7, and Table A3). Third, when further considering first-author information to proxy three control variables, whether the first author's institution is among the top 100 universities, whether the first author works in an

[4] While Tables 2 and 3 summarize related data, Table 2 highlights dimension-level manual agreement, whereas Table 3 displays the overall scores comparison between human evaluators and Gemini 2.0 Flash.

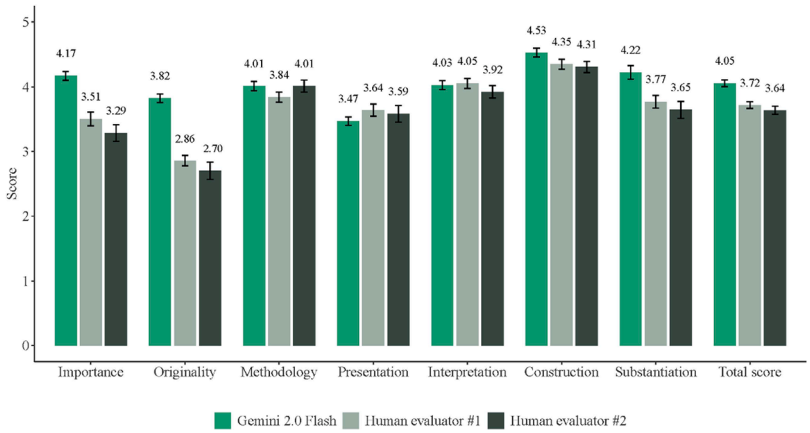

Fig. 2. The item score for human evaluators and Gemini 2.0 Flash. Error bars represent 95% confidence intervals. This figure is plotted based on a subsample of 300 PRRs.

Table 2
Statistical and agreement analysis of item scores evaluated by human evaluators.

RQI items	Mean (SD)		Difference (95% CI)	Agreement
	Evaluators #1	Evaluators #2		
Importance	3.51 (0.93)	3.29 (1.14)	0.22 (0.05, 0.38)	0.99
Originality	2.86 (0.71)	2.70 (1.17)	0.16 (0.00, 0.31)	0.99
Methodology	3.84 (0.67)	4.01 (0.83)	-0.17 (-0.28, -0.05)	0.99
Presentation	3.64 (0.83)	3.59 (1.14)	0.05 (-0.11, 0.21)	0.99
Interpretation	4.05 (0.67)	3.92 (0.84)	0.13 (0.01, 0.25)	0.99
Construction	4.35 (0.68)	4.31 (0.76)	0.04 (-0.07, 0.16)	0.99
Substantiation	3.77 (0.87)	3.65 (1.15)	0.12 (-0.04, 0.29)	0.99

Notes: This analysis is based on a subsample of 300 PRRs.

Table 3
Agreement between human evaluators and Gemini 2.0 Flash.

Evaluators	Mean (SD)	Agreement (1)
(1) Gemini 2.0 Flash	4.05 (0.44)	/
(2) Evaluator 1	3.72 (0.46)	0.86
(3) Evaluator 2	3.64 (0.56)	0.79

Notes: Results are based on the average overall score of a subsample of 300 PRRs.

English-speaking country, and whether the reviewers' academic status is higher than the first author's. This is because, although both first authors and last corresponding authors, on average, make substantial contributions to the work, first authors generally contribute more than last corresponding authors (Corrêa et al., 2017). The re-estimated results reported in Table A4 were comparable to the main results. This reinforces that constructing these controls based on either the first author or the last corresponding author did not affect the study's findings. Overall, all robustness checks suggest that the estimation results are solid.

5. Discussion

Given the substantial increase in the volume of scientific publications, it is crucial to examine whether peer review effectively safeguards research quality, and therefore enhancing the scientific impact. Utilizing the latest version of Google's LLMs, Gemini 2.0 Flash, I automatically assess the quality of peer reviews in biomedical papers using the RQI and analyzed the number of citations within the first three years after publication as a proxy for scientific impact. Findings indicate that high-quality peer reviews are positively associated with the subsequent scientific influence of the reviewed papers. Specifically, the estimated coefficient is positive and statistically significant (5.440, $p < 0.01$), suggesting that a one-point increase in review score, reflecting the peer review quality assessed by Gemini 2.0 Flash, corresponds to an average of 5.440 additional citations. When using both average citations and the "year-normalised" citations as dependent variables, the positive effect remain persist. The results imply that reviewers who provide high-quality feedback may contribute to improving the quality of submissions, thereby increasing their recognition within the scientific community and ultimately leading to higher citation counts.

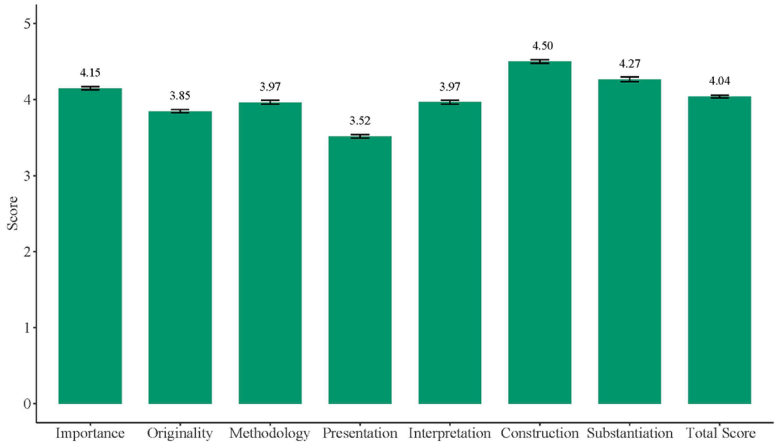

Fig. 3. The score distribution of Gemini 2.0 Flash. Error bars represent 95% confidence intervals. This figure is plotted based on a total of 2,580 PRRs.

Table 4
Linear regression estimation results and robustness checks.

	Main result	Robustness checks		
	(1) Citations	(2) ln(Citations)	(3) Citations excluding Top 1%	(4) Nb-reg
Review quality	5.440***	0.166***	2.813***	0.181***
	(1.643)	(0.038)	(0.820)	(0.048)
Observations	2,580	2,580	2,554	2,580
R-squared	0.049	0.097	0.063	/
Controlled Variables	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y

Notes: Parentheses reported the robust standard errors. Control variable include: (1) title length; (2) number of pages; (3) abstract length; (4) number of references; (5) number of authors; (6) whether the institution of last corresponding author is top 100 universities; (7) whether the last corresponding author works in English speaking countries; and (8) whether the academic status of reviewers is higher than the last corresponding author. The results remain robust even when year fixed effects are excluded. Detailed control variable estimates are presented in Table A1. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

This study benefits from the rapid advancements in the development of LLMs and their growing academic attention in the evaluation of tasks related to peer review. Existing research has assessed the quality of peer review feedback generated by LLMs, such as GPT-4, and compared it with feedback provided by human evaluators (Liang et al., 2024). The findings indicate that GPT-4's feedback significantly overlaps with that of human reviewers. Notably, in the case of ICLR papers, the overlap reached as high as 77.18%, providing fundamental evidence for the LLMs' potential in automatically assessing peer review quality. Different from their research design, this study builds on the use of most recent LLM of Gemini 2.0 Flash, which exhibit promising capabilities in answering complex questions and offering detailed explanations of the "thinking" process. Additionally, this study exclusively employs the RQI, a widely recognized tool in assessing the quality of peer reviews within the biomedical research domain. These methodological choices enhance the credibility of the evidence presented in this study.

To further strengthen the claim that the review scores automatically evaluated by Gemini 2.0 Flash are reliable and comparable to those of human evaluators, I conduct an analysis by using a subsample of 300 PPRs. This analysis compares the evaluation performance of Gemini 2.0 Flash with that of two human evaluators. The human evaluators undergo thorough training to ensure they are qualified to perform the evaluation tasks. The results of the analysis reveal excellent agreement between the automatic and human evaluations of peer review quality. The observed agreement is chose because of the narrow range of Likert-scale scores, where kappa or correlation analysis cannot capture the substantial consistency of a difference of 1 (Rothwell & Martyn, 2000).The finding suggests that LLMs, such as Gemini 2.0 Flash, have the potential to alleviate the time-consuming process of evaluating peer review quality (Thelwall, 2024a), and exhibit potential that LLMs might be useful for journal editors, conference organizers, or other parties interested in assessing the quality of peer reviews. However, the findings should be regarded as an initial effort in assessing the performance of Gemini 2.0 Flash in peer review quality evaluation. Further research and development are necessary to fully establish its effectiveness and reliability in this domain, given the ongoing controversy and ethical concerns surrounding the use of LLMs for automated peer review related tasks.

Despite emphasizing the importance of rigorous and high-quality peer reviews as a key factor in enhancing the scientific impact of published research, it is important to acknowledge that other factors, such as the inherent novelty or significance of the research, may also influence scientific impact. Research quality remains a multifaceted concept, and there is still no consensus on how to define or

measure it comprehensively (Aksnes et al., 2019). While indicators such as novelty, disruption (and its variants), and other metrics have been proposed to assess research quality (Arts et al., 2021; Funk & Owen-Smith, 2017; Leibel & Bornmann, 2024; Wu & Wu, 2019; Wu & Yan, 2019; Wu et al., 2019), these are often citation-based and suffer from the inherent limitations of citation alone is not sufficient to measure the true novelty of a work. Evaluating the quality of peer reviews provides a valuable alternative perspective for assessing research quality. However, it is important to note that existing peer review quality evaluation scales remain largely confined to the fields of life sciences and medicine, underscoring the need for broader and interdisciplinary approaches (Superchi et al., 2019).

Limitations

This study inevitably has limitations. First and most importantly, LLMs raise significant concerns regarding bias (Ye et al., 2024), as they are trained on vast and varied datasets source where intergroup prejudice is often pervasive. Therefore, the growing reliance on LLMs to replace human judgment in assessing peer review quality requires more in-depth analysis (Bail, 2024; (Goldberg et al., 2024; Liu & Shah, 2023). Second, due to resource limitations, only two types of prompts (few-shot prompt and zero-shot prompt) are utilized to guide LLMs in generating responses (Reynolds & McDonell, 2021). Fine-tuning prompt techniques to enhance the accuracy and efficiency of LLM performance warrant further exploration (Sahoo et al., 2024). Also, when without resource constraints, using a relatively larger sample size to verify the conclusions of this study is recommended for future research. Third, the review scores assessed by Gemini 2.0 Flash are based on a 5-point Likert scale. The limited range of this scale may contribute to the lack of variation observed across the seven items. The potential impact of this constraint warrants further investigation. Finally, this analysis is limited to PPRs in the life sciences and biomedical domains by using the well-established RQI tool in a unique journal. It is important to highlight that other factor, such as research significance, may also influence scientific impact but were not explicitly included as review quality items in the current RQI. Generalizing the findings to a broader range of research disciplines with additional items to address review quality is needed.

6. Conclusions

This study utilizes Google's most recent LLM, Gemini 2.0 Flash, to automate the evaluation of peer review quality for biomedical research using the widely recognized RQI tool. And the relationship between automatically assessed review scores and scientific impact is examined. The results demonstrate that Gemini 2.0 Flash provides reliable and comparable evaluations to those of human reviewers for PPRs, highlighting its potential for large-scale PPRs evaluation. Further linear regression analysis reveals a positive association between review scores automatically assessed by Gemini 2.0 Flash and subsequent citations. Specifically, a one-point increase in the review score corresponds to an average increase of approximately five citations. Several robustness checks, including transforming citation counts into their logarithmic form, excluding papers within the top 1% of citations, and employing negative binomial regression model, confirm the robustness of the estimated results. These findings underscore the promise of LLMs in the automatic assessment of peer review quality; however, further efforts are needed to ensure their reliability and enhance their application across diverse research domains.

Declaration of generative AI and AI-assisted technologies in the writing process

During the first round of revision, the author used ChatGPT to proofread and improve the readability and language quality of certain sections of the manuscript. After using this tool, the authors reviewed and edited the content and take full responsibility for the content of the published article.

Data availability statement

The data and code for this study are available upon request.

CRediT authorship contribution statement

Zhuanlan Sun: Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

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Appendix

Example 1: High-Quality Review

```
### Example 1: High-Quality Review
**Sample Review**: "This study addresses a significant gap in understanding neural plasticity mechanisms. The research question is well-motivated and builds upon recent advances in the field. The methodology is rigorous, employing both in vitro and in vivo approaches with appropriate controls. However, the sample size for the behavioral experiments could be larger (n=8 per group). The results are clearly presented with well-designed figures, though Figure 3b could benefit from error bars. The discussion appropriately acknowledges limitations and suggests concrete future directions. The writing is clear and well-organized throughout."

**Evaluation**:
```json
{
  "score_explanation": "This review demonstrates excellent analytical depth across all dimensions, providing specific methodological feedback and constructive suggestions. The reviewer maintained objectivity while offering detailed, evidence-based comments.",
  "individual_scores": {
    "research_importance": 4,
    "originality": 4,
    "methodology": 5,
    "presentation": 4,
    "results_interpretation": 4,
    "constructive_feedback": 5,
    "evidence_support": 5
  },
  "total_score": 4.4,
  "detailed_comments": {
    "strengths": ["Specific methodological analysis", "Concrete suggestions for improvement", "Balanced assessment of strengths and weaknesses"],
    "areas_for_improvement": ["Could elaborate more on significance implications"],
    "single_blind_considerations": ["Maintained objectivity despite likely familiarity with author's work"]
  }
}
```
```

Fig. A1. High-quality review example illustrated in the few-shot prompt.

Example 2: Medium-Quality Review

```
### Example 2: Medium-Quality Review
**Sample Review**: "The paper investigates machine learning applications in healthcare prediction. The topic is relevant but not particularly novel. The methods seem appropriate though more details on validation would be helpful. Results are presented clearly. Some figures could be improved. Overall the work is solid but incremental."

**Evaluation**:
```json
{
  "score_explanation": "This review covers basic elements but lacks depth in critical analysis and specific guidance. While generally fair, it provides limited actionable feedback for authors.",
  "individual_scores": {
    "research_importance": 3,
    "originality": 2,
    "methodology": 3,
    "presentation": 3,
    "results_interpretation": 3,
    "constructive_feedback": 2,
    "evidence_support": 2
  },
  "total_score": 2.6,
  "detailed_comments": {
    "strengths": ["Covers main review categories", "Generally accurate assessment"],
    "areas_for_improvement": ["Lacks specific examples", "Minimal constructive suggestions", "Superficial analysis"],
    "single_blind_considerations": ["No apparent bias issues detected"]
  }
}
```
```

Fig. A2. Medium-quality review example illustrated in the few-shot prompt.

Example 3: Low-Quality Review

```
### Example 3: Low-Quality Review
**Sample Review**: "This work is not innovative and the methods are questionable. The writing needs significant improvement. I recommend rejection."

**Evaluation**:
```json
{
  "score_explanation": "This review fails to provide substantive analysis or constructive feedback, offering only superficial criticism without supporting evidence or specific guidance for improvement.",
  "individual_scores": {
    "research_importance": 1,
    "originality": 1,
    "methodology": 1,
    "presentation": 1,
    "results_interpretation": 1,
    "constructive_feedback": 1,
    "evidence_support": 1
  },
  "total_score": 1.0,
  "detailed_comments": {
    "strengths": ["Brief and to the point"],
    "areas_for_improvement": ["No specific evidence provided", "Lacks constructive suggestions", "Minimal analysis across all dimensions", "Does not help authors improve their work"],
    "single_blind_considerations": ["Possible bias influence cannot be ruled out due to lack of substantiation"]
  }
}
```
```

Fig. A3. Low-quality review example illustrated in the few-shot prompt.

Details of the Instruction of RQI Items

Evaluate these specific aspects:
1. Research Question Importance: Assessment of how well the reviewer addressed the significance of the research question
2. Question Originality: Evaluation of how well the reviewer assessed the novelty and originality of the research
3. Methodological Analysis: Assessment of how thoroughly the reviewer examined the strengths and weaknesses of the methodology
4. Presentation Review: Evaluation of the reviewer's assessment of writing quality, organization, and visual elements
5. Results Interpretation: Assessment of how well the reviewer evaluated the interpretation and analysis of results
6. Constructive Feedback: Evaluation of the extent to which the reviewer provided helpful and actionable comments
7. Evidence-Based Commentary: Assessment of how well the reviewer supported their comments with specific evidence and maintained objectivity despite knowing author identity

Fig. A4. Details of the Instruction of RQI items for both few-shot and zero-shot prompts.

Details of the Query for Task Evaluation

```
## Your Evaluation Task

Please provide your evaluation in the following JSON format:

{
  "score_explanation": "<2-3 sentences providing a balanced assessment of the review's overall quality, highlighting both general strengths and key areas needing improvement, considering the single-blind context>",
  "individual_scores": {
    "research_importance": <score>,
    "originality": <score>,
    "methodology": <score>,
    "presentation": <score>,
    "results_interpretation": <score>,
    "constructive_feedback": <score>,
    "evidence_support": <score>
  },
  "total_score": <mean_score>,
  "detailed_comments": {
    "strengths": ["<strength_1>", "<strength_2>", ...],
    "areas_for_improvement": ["<area_1>", "<area_2>", ...],
    "single_blind_considerations": ["<consideration_1>", "<consideration_2>", ...]
  }
}
```

Fig. A5. Details of the query for task evaluation for both few-shot and zero-shot prompts.

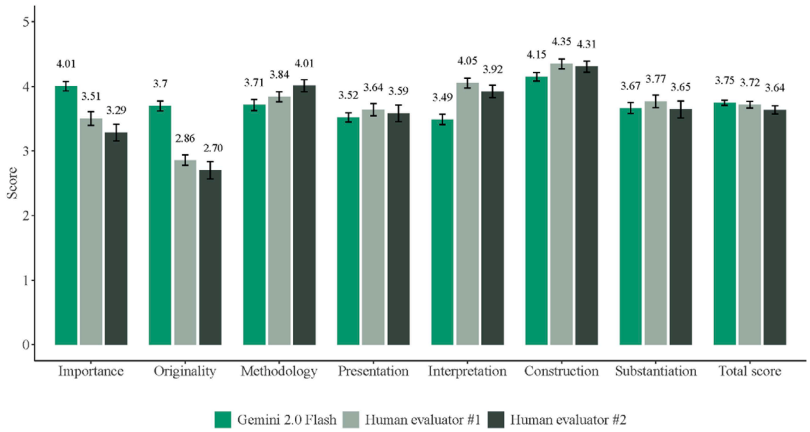

Fig. A6. The item score of zero-few prompt for human evaluators and Gemini 2.0 Flash. Error bars represent 95% confidence intervals. This figure is plotted based on a subsample of 300 PRRs.

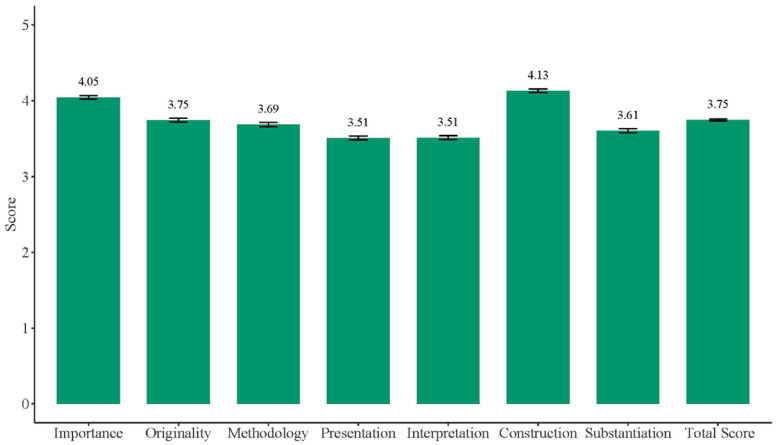

Fig. A7. The zero-few prompt score distribution of Gemini 2.0 Flash. Error bars represent 95% confidence intervals. This figure is plotted based on a total of 2,580 PRRs.

Table A1
Linear regression estimation results with detailed control variable estimates.

	Main result	Robustness checks		
	(1) Citations	(2) ln(Citations)	(3) Citations excluding Top 1%	(4) Nb-reg
Review quality	5.440***	0.166***	2.813***	0.181***
	(1.643)	(0.038)	(0.820)	(0.048)
Title lengths	-0.495**	-0.012*	-0.288**	-0.018***
	(0.206)	(0.006)	(0.114)	(0.006)
Number of pages	0.281*	0.002	0.057	0.005
	(0.150)	(0.003)	(0.066)	(0.004)
Abstract lengths	0.178*	0.002*	0.037	0.002
	(0.096)	(0.001)	(0.030)	(0.001)
Number of references	-0.000	0.003***	0.047***	0.001
	(0.035)	(0.001)	(0.016)	(0.001)
Number of authors	0.066	0.002*	0.035*	0.016
	(0.051)	(0.001)	(0.019)	(0.016)
Affiliation of a top university	-0.983	0.010	-0.001	-0.007
	(1.239)	(0.040)	(0.796)	(0.042)
Affiliation of English-speaking country	1.517	0.069*	1.104	0.055
	(1.229)	(0.036)	(0.718)	(0.044)
Academic status of reviewer	-2.539*	-0.083**	-1.522**	-0.109**
	(1.484)	(0.037)	(0.751)	(0.047)
Year fixed effects	Y	Y	Y	Y
Observations	2,580	2,580	2,554	2,580
R-squared	0.049	0.097	0.063	/

Notes: Parentheses reported the robust standard errors. Control variable include: (1) title length; (2) number of pages; (3) abstract length; (4) number of references; (5) number of authors; (6) whether the institution of last corresponding author is top 100 universities; (7) whether the last corresponding author works in English speaking countries; and (8) whether the academic status of reviewers is higher than the last corresponding author. The results remain robust even when year fixed effects are excluded. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A2
Estimation results and robustness checks based on average and year-normalised citations.

	Main result	Robustness checks		
	(1) Citations	(2) ln(Citations)	(3) Citations excluding Top 1%	(4) Nb-reg
Panel A: Average citations as dependent variable				
Review quality	1.497***	0.162***	0.932***	0.168***
	(0.392)	(0.037)	(0.274)	(0.043)
Observations	2,580	2,580	2,554	2,580
R-squared	0.048	0.098	0.063	/
Controlled Variables	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y
Panel B: Year-normalised citations as dependent variable				
Review quality	0.614***	0.152***	0.319***	0.170***
	(0.211)	(0.039)	(0.111)	(0.055)
Observations	2,580	2,580	2,554	2,580
R-squared	0.027	0.052	0.024	/
Controlled Variables	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y

Notes: Parentheses reported the robust standard errors. Control variable include: (1) title length; (2) number of pages; (3) abstract length; (4) number of references; (5) number of authors; (6) whether the institution of last corresponding author is top 100 universities; (7) whether the last corresponding author works in English speaking countries; and (8) whether the academic status of reviewers is higher than the last corresponding author. The results remain robust even when year fixed effects are excluded. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3
Estimation results and robustness checks based on zero-few prompt.

	Main result	Robustness checks		
	(1) Citations	(2) ln(Citations)	(3) Citations excluding Top 1%	(4) Nb-reg
Review quality	4.391**	0.095**	1.843*	0.135**
	(2.021)	(0.048)	(1.042)	(0.059)
Observations	2,580	2,580	2,554	2,580
R-squared	0.046	0.093	0.060	/
Controlled Variables	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y

Notes: Parentheses reported the robust standard errors. Control variable include: (1) title length; (2) number of pages; (3) abstract length; (4) number of references; (5) number of authors; (6) whether the institution of last corresponding author is top 100 universities; (7) whether the last corresponding author works in English speaking countries; and (8) whether the academic status of reviewers is higher than the last corresponding author. The results remain robust even when year fixed effects are excluded. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4
Estimation results of using first author information to proxy control variables.

	Main result	Robustness checks		
	(1) Citations	(2) ln(Citations)	(3) Citations excluding Top 1%	(4) Nb-reg
Review quality	5.317***	0.162***	2.758***	0.173***
	(1.600)	(0.038)	(0.824)	(0.046)
Title lengths	-0.475**	-0.011*	-0.286**	-0.017***
	(0.198)	(0.006)	(0.114)	(0.006)
Number of pages	0.281*	0.002	0.060	0.005
	(0.150)	(0.003)	(0.067)	(0.004)
Abstract lengths	0.179*	0.002**	0.038	0.002
	(0.096)	(0.001)	(0.030)	(0.001)
Number of references	-0.003	0.003***	0.046***	0.001
	(0.036)	(0.001)	(0.016)	(0.001)
Number of authors	0.064	0.002*	0.034*	0.015
	(0.051)	(0.001)	(0.019)	(0.016)
Affiliation of a top university	-0.405	0.032	0.302	0.005
	(1.276)	(0.040)	(0.795)	(0.042)
Affiliation of English-speaking country	0.168	0.036	0.814	0.027
	(1.107)	(0.036)	(0.717)	(0.045)

(*continued on next page*)

Table A4 (*continued*)

	Main result	Robustness checks		
	(1) Citations	(2) ln(Citations)	(3) Citations excluding Top 1%	(4) Nb-reg
Academic status of reviewer	-7.330	-0.037	-0.528	-0.210
	(6.702)	(0.080)	(1.683)	(0.155)
Year fixed effects	Y	Y	Y	Y
Observations	2,580	2,580	2,554	2,580
R-squared	0.049	0.095	0.062	/

Notes: Parentheses reported the robust standard errors. Control variable include: (1) title length; (2) number of pages; (3) abstract length; (4) number of references; (5) number of authors; (6) whether the institution of the first author is top 100 universities; (7) whether the first author works in English speaking countries; and (8) whether the academic status of reviewers is higher than the first author. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

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